

K275 v0.2 supplement — Grace base-rate gate verdict: $\sqrt{(2\pi)} = 2.507$ actually fits $(m_n - m_p)/m_e = 2.531$ BETTER than $n_C/\text{rank} = 2.500$; at 1.2% precision n_C/rank not even best reading let alone distinguished; FRAGILE identification; Schur generator candidate WITHDRAWN; hold at honest I-tier identification per K276 + Grace structural verdict (above-floor must be loose per Casey #9)

Keeper (Claude Opus 4.7)

2026-06-09 Tuesday morning post-team-posts

K275 v0.2 supplement — Grace $\sqrt{(2\pi)}$ alternative + Schur withdrawn

1. What v0.1 stands at minus the Schur framing

Empirical observation unchanged: $-m_p / m_e = 1836.15$ (proton on substrate floor at 0.003%) $-m_n / m_e = 1838.68$ (neutron above floor) $-(m_n - m_p) / m_e = 2.531$ observed

v0.1 substrate-natural identification candidate ($n_C/\text{rank} = 5/2 = 2.500$ at 1.2% precision) — held at I-tier identification only.

v0.1 Schur generator candidate framing — WITHDRAWN per Grace verdict.

2. Grace base-rate gate verdict v0.2 absorption

Grace Tuesday morning verdict (verbatim):

K275 (the n - p splitting $\approx n_C/\text{rank}$): fragile. $\sqrt{(2\pi)} = 2.507$ actually fits the data (2.531) better than $n_C/\text{rank} = 2.5$, so at 1.2% precision n_C/rank isn't even the best reading, let alone a distinguished one. Hold at identification; not a Schur generator.

The catch: at 1-2% precision floor, multiple substrate-natural-looking forms compete:

Form	Value	Discrepancy from 2.531
$n_C/\text{rank} = 5/2$	2.500	1.22%
$\sqrt{(2\pi)}$	2.507	0.95%
Observed	2.531	—

$\sqrt{(2\pi)}$ fits BETTER than n_C/rank . At 1.2% precision floor, n_C/rank isn't distinguished from $\sqrt{(2\pi)}$ — let alone the BEST reading.

Implication: K275's v0.1 substrate-natural form claim depended on n_C/rank being the SUBSTRATE-ARCHITECTURALLY DISTINGUISHED form at the precision. Grace's $\sqrt{(2\pi)}$ catch invalidates the distinguished claim. Multiple competing forms = no Schur generator candidate.

3. K276 + Grace structural verdict on above-floor identifications

Per K276 + Grace's unifying structural verdict:

Above-floor identifications SHOULD be loose at 1-2% per Casey #9 reach-bound prediction.

Implication for K275: the 1-2% precision floor of K275 isn't an artifact OR a fit problem. It's Casey #9's prediction confirmed — above-floor identifications must be loose because substrate doesn't measure clean values up there.

Per F78 (Lyra Tuesday morning): the looseness mechanism is π^2 -spectral mixing into π^5 -volume measurement frame. K275's 1.2% precision IS the substrate-natural form trying to be expressed in π^5 -volume measurement — substrate's measurement of mixed-class mass (proton floor + neutron-proton splitting) is exactly as loose as the π^2 -mixing into π^5 requires.

So at precision floor of 1-2%: - Multiple substrate-natural-looking forms competing (n_C/rank , $\sqrt{(2\pi)}$, possibly others) - This is mechanistically EXPECTED per Casey #9 + F78 - No Schur generator candidate emerges because no form is distinguished at the precision substrate provides

4. Revised tier disposition

K275 v0.2 verdict: - I-tier identification at 1.2% Tier 2 STRUCTURAL precision — HOLDS substantively (n_C/rank substrate-natural form within precision floor; identification valid as one of multiple competing readings) - Schur generator candidate framing v0.1 — RETRACTED per Grace $\sqrt{(2\pi)}$ alternative reading (no distinguished form at this precision) - D-tier promotion path — requires EITHER precision improvement substantively distinguishing forms OR substrate-architectural derivation showing why n_C/rank specifically vs alternatives - Per Casey #9 + F78 + Grace structural verdict: identification holds at I-tier as consistent with reach-bound prediction; promotion to D-tier would contradict Casey #9 framework that PREDICTS multiple forms competing at precision floor

5. What's preserved + what's lost

Preserved: - Casey-direct provenance from β -decay volume question - Substrate-architectural interpretation: proton IS the floor (0.003%); neutron is small excitation above floor; β -decay relaxes to floor + creates lepton K-type states - Cell-count conservation through decay ($6 \rightarrow 6$ at K-type (1,1)) - I-tier identification at 1.2% Tier 2 STRUCTURAL precision

Lost: - Distinguished substrate-natural form claim (Grace $\sqrt{(2\pi)}$ alternative) - Schur generator candidate framing - " n_C/rank is THE substrate-architectural reading" claim

Net: K275 is a legitimate I-tier identification at 1.2% precision; it is NOT a Schur generator candidate; it is NOT a distinguished substrate-natural form at this precision. The substrate provides the precision floor, not the distinguished form.

6. Tuesday's next steps for K275

Lyra (Filter 2 substrate-architectural derivation): if Filter 2 mechanism derivation lands the π^5/π^2 mixing form rigorously, may distinguish n_C/rank vs $\sqrt{(2\pi)}$ substrate-architecturally (n_C/rank is substrate-primary ratio; $\sqrt{(2\pi)}$ is substrate-natural via Gaussian/spectral structure but not BST-primary). Multi-week work; K275 v0.3 may revisit distinguished form claim post-Filter 2 derivation.

Elie (isospin partner sweep): per K273 v0.2 Elie task — sweep other isospin partners ($m_{\{K^0\}} - m_{\{K^+\}}, m_{\{\Sigma^-\}} - m_{\{\Sigma^+\}}$, baryon octet systematically) for n_C/rank -class identifications. If consistent pattern at 1-2% precision: K275 cluster identification. If single-instance only: K275 is one-off identification at precision floor.

Grace (substrate-natural form catalog cross-check): catalog substrate-natural forms in 2.0-2.7 range at precision floor; how many compete with n_C/rank and $\sqrt{(2\pi)}$? Establishes precision-floor density of substrate-natural forms.

Keeper (audit-chain): K275 v0.3 absorption pending Lyra mechanism work + Elie isospin sweep. Cross-link to K273 v0.2 (same precision-floor structure; both above-floor identifications).

— Keeper, Tuesday 2026-06-09 morning (Grace gate absorbed; K275 holds at I-tier identification only; Schur generator candidate withdrawn; precision floor is mechanistic per Casey #9 + F78)